Physics-Guided Temporal Decomposition and Chronosequence Attention for Climate-Driven Forest Growth Prediction

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***Abstract*—Accurate forest growth prediction is important for understanding terrestrial carbon dynamics, ecosystem productivity, and the impact of climate variability on vegetation development. Traditional statistical models may struggle to represent nonlinear relationships among climate, forest age, and biomass accumulation. This paper presents a physics-guided temporal decomposition framework for predicting forest height and above-ground biomass (AGB) using historical climate drivers and forest age information.**

The proposed framework combines a logarithmic ecological growth prior, termed EcoPrior, with two complementary temporal branches. A Conv1D and bidirectional gated recurrent unit (BiGRU) branch captures long-term climate trends, while a one-dimensional convolutional neural network captures monthly seasonal variability. A stand chronosequence attention mechanism integrates information across fifteen initial forest ages to represent age-dependent interactions. The neural network predicts residual corrections to the ecological baseline. Physics-based constraints, ensemble averaging, residual blending, and output calibration are subsequently applied during inference.

The reported implementation uses four random seeds and ten cross-validation folds, producing forty neural models. The final inference configuration uses a residual blending coefficient of 0.350, specialized clamping for tundra and barren-ground conditions, and five-pass mean alignment. The reported calibration targets are 0.899911 for height and 0.923806 for AGB. The complete prediction pipeline achieved a champion competition score of 0.072.

***Index Terms*—Forest Growth Prediction, Above-Ground Biomass, Climate Modeling, Physics-Guided Machine Learning, BiGRU, Temporal Decomposition, Chronosequence Attention, Deep Learning, Carbon Dynamics**

I. Introduction

Forests play an important role in the global carbon cycle by absorbing atmospheric carbon dioxide and storing carbon in vegetation and soil. Predicting forest growth under changing climatic conditions is essential for ecosystem monitoring, carbon estimation, and climate-impact assessment.

Forest development depends on multiple interacting factors, including temperature, precipitation, soil characteristics, vegetation age, and competition for environmental resources. These relationships are nonlinear and may vary across temporal scales and ecological regions.

Traditional forest growth models use empirical growth curves or process-based equations to estimate vegetation development. Although these methods provide useful ecological structure, they may not fully capture complex interactions in high-dimensional environmental datasets.

Deep learning provides an alternative by learning nonlinear relationships directly from data. However, unconstrained neural predictions can produce ecologically implausible results, particularly in regions with little vegetation. Furthermore, a single temporal representation may fail to distinguish long-term climate trends from short-term seasonal variability.

This paper presents a hybrid framework that combines ecological prior knowledge with temporal deep learning and physics-aware post-processing.

The principal contributions are:

1) A logarithmic ecological growth prior, EcoPrior, for constructing baseline vegetation predictions.
2) A dual-branch temporal architecture that separates long-term climate trends from monthly weather variability.
3) A stand chronosequence attention mechanism that models relationships across fifteen initial forest ages.
4) A residual neural prediction strategy that corrects the physical baseline instead of relying exclusively on unconstrained absolute predictions.
5) A physics-aware inference pipeline incorporating land-cover clamping, ensemble averaging, residual blending, and mean calibration.

II. Related Work

A. Forest Growth Modeling

Forest growth modeling includes empirical growth equations, process-based ecosystem models, and statistical prediction methods. Empirical models describe age-dependent changes in vegetation, while process-based models represent ecological processes involving productivity, respiration, and carbon allocation.

A logarithmic growth transformation can represent diminishing increments with age. However, actual forest development also depends on environmental conditions, disturbance, and ecosystem-specific characteristics.

The proposed approach uses an ecological prior as a baseline and allows a neural network to learn corrections from climate and age information.

B. Deep Learning for Temporal Data

Convolutional neural networks can extract local patterns from sequential data, while recurrent neural networks represent temporal dependencies.

Gated recurrent units regulate information flow through update and reset gates and are suitable for sequential representation learning [1].

The combination of convolutional and recurrent layers provides a way to learn local temporal features and long-range dependencies. In the proposed framework, these operations are applied to long-term climate trends.

C. Attention Mechanisms

Attention mechanisms assign different weights to different elements of an input representation. Transformer-based attention provides a general framework for learning relationships between sequence elements [2].

In forest modeling, attention across different age states can learn age-dependent relationships relevant to vegetation development. Such weights represent learned statistical relationships and should not automatically be interpreted as direct measurements of biological competition.

D. Physics-Guided Machine Learning

Physics-guided machine learning combines domain knowledge with data-driven models. Karpatne et al. discussed the use of scientific knowledge in Earth systems [3], [4].

The proposed approach incorporates domain knowledge through a physical growth prior, residual learning, and output constraints. These components aim to reduce reliance on unrestricted neural predictions.

E. Research Gap

The proposed framework addresses the following questions:

- Can separate temporal branches represent long-term climate trends and seasonal variability?
- Can attention across forest ages learn useful chronosequence relationships?
- Can residual prediction and physical constraints reduce unrealistic ecological estimates?
- Can ensemble averaging and output calibration provide a stable final prediction pipeline?

Controlled ablation experiments are required to quantify the independent contribution of each component.

III. Methodology

A. Problem Formulation

Let the historical climate input for site i be

$$X_i \in \mathbb{R}^{40\times 12\times F}, \tag{1}$$

where 40 denotes the number of historical years, 12 denotes the number of months per year, and F represents the number of climate features.

The monthly climate sequence therefore contains

$$T = 40 \times 12 = 480 \tag{2}$$

monthly observations.

The model considers fifteen initial forest ages:

$$\mathcal{A} = \{a_1, a_2, \ldots, a_{15}\}. \tag{3}$$

For each site and age configuration, the model predicts forest height and above-ground biomass:

$$\hat{Y}_{i,a} = \begin{bmatrix} \hat{H}_{i,a} \\ \widehat{AGB}_{i,a} \end{bmatrix}. \tag{4}$$

The training objective additionally includes auxiliary ecological variables such as soil, leaf area index, gross primary productivity, net primary productivity, and heterotrophic respiration.

B. Overall Architecture

The proposed pipeline combines five components:

1) EcoPrior physical growth estimation.
2) Long-term climate trend encoding.
3) Monthly seasonal weather encoding.
4) Stand chronosequence attention and residual prediction.
5) Physics-aware clamping and calibration.

Fig. 1 illustrates the architecture.

C. Data Representation and Preprocessing

The input consists of historical climate sequences and initial ecological states.

A common feature-wise standardization is

$$\tilde{x}_j = \frac{x_j - \mu_j}{\sigma_j + \epsilon}, \tag{5}$$

where μ_j and σ_j represent the mean and standard deviation of feature j, and ϵ is a numerical stability constant.

The model separates the climate information into long-term and seasonal representations:

$$X_{\text{trend}} = f_{\text{trend}}(X), \tag{6}$$

$$X_{\text{season}} = f_{\text{season}}(X). \tag{7}$$

The trend representation captures multi-decadal climate variation, while the seasonal representation captures shorter-term monthly fluctuations.

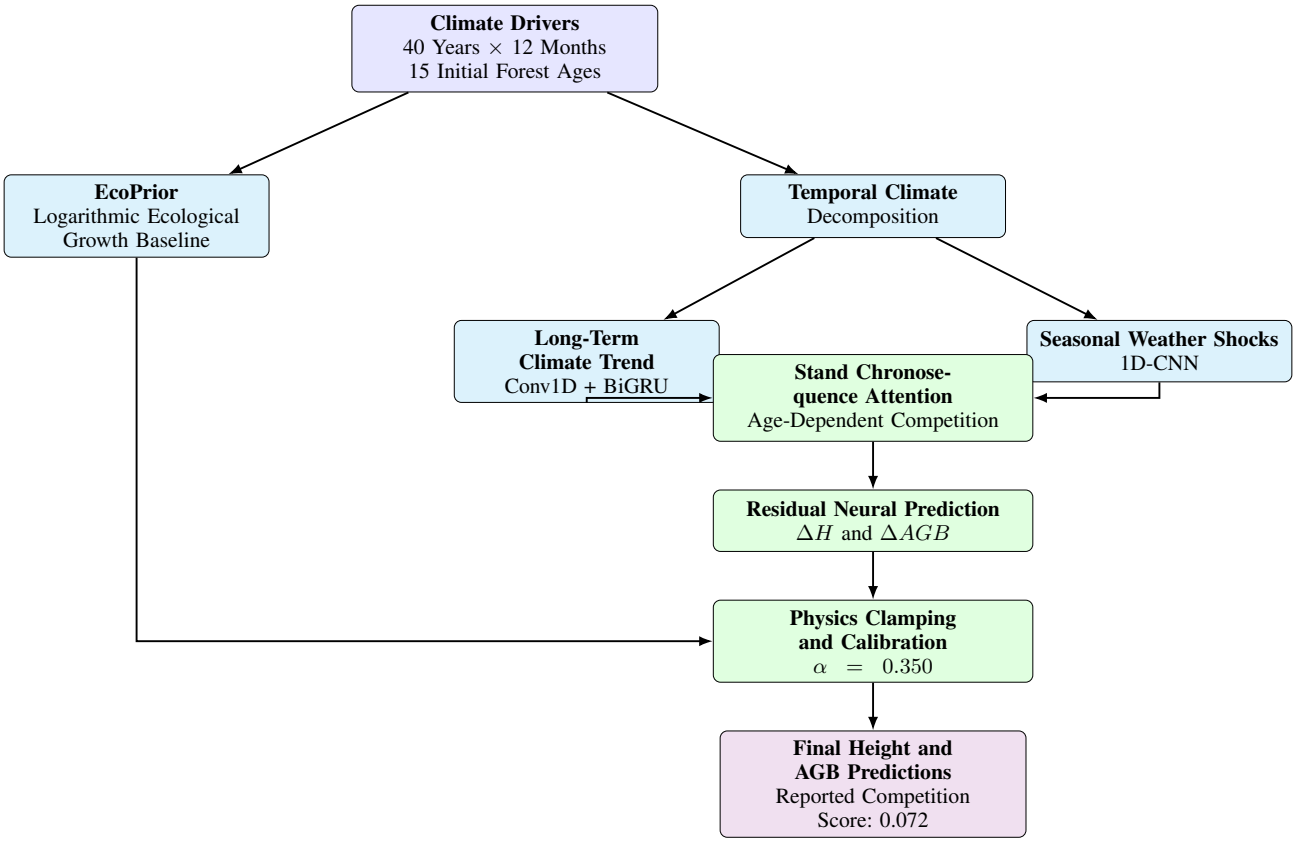


Fig. 1. Overview of the proposed physics-guided temporal decomposition and chronosequence attention framework.

D. EcoPrior: Physical Growth Baseline

Forest growth depends on stand age, and growth increments may diminish as forests mature.

The proposed EcoPrior uses a logarithmic age transformation:

$$g(a) = \log(a + 1). \tag{8}$$

Let the initial ecological state be

$$Y_0(a) = [H_0(a), AGB_0(a), \ldots]. \tag{9}$$

The physical baseline at a future age is represented by

$$Y_{\text{phys}}(a + \Delta t) = f_{\text{EcoPrior}}\left(Y_0, \log(a + \Delta t + 1)\right). \tag{10}$$

This formulation provides an ecological reference for the neural model. The prior supplies the baseline prediction, while the neural network estimates data-dependent residuals.

E. Long-Term Climate Trend Encoder

The long-term branch uses a one-dimensional convolutional encoder followed by a bidirectional gated recurrent unit.

The convolutional representation is

$$Z_t = \phi(W_c * X_{\text{trend}} + b_c), \tag{11}$$

where $*$ denotes convolution and ϕ is a nonlinear activation function.

The forward and backward recurrent representations are

$$\overrightarrow{h}_t = GRU_f(Z_t, \overrightarrow{h}_{t-1}), \tag{12}$$

$$\overleftarrow{h}_t = GRU_b(Z_t, \overleftarrow{h}_{t+1}). \tag{13}$$

They are concatenated to obtain

$$h_t = [\overrightarrow{h}_t; \overleftarrow{h}_t]. \tag{14}$$

The resulting representation summarizes long-term climate dependencies over the observed historical window.

F. Seasonal Weather Shock Encoder

Long-term climate trends alone may not capture short-duration environmental disturbances. Monthly temperature, precipitation, and other weather variations may influence vegetation productivity.

A separate one-dimensional convolutional network processes the seasonal representation:

$$Z_{\text{season}} = \phi\left(Conv1D(X_{\text{season}})\right). \tag{15}$$

The extracted seasonal features are aggregated to obtain

$$h_{\text{season}} = Pool(Z_{\text{season}}). \tag{16}$$

The trend and seasonal representations are combined as

$$h_{\text{climate}} = [h_{\text{trend}}; h_{\text{season}}]. \tag{17}$$

This combined representation provides information about both long-term climatic conditions and monthly variability.

G. Stand Chronosequence Attention

The framework considers fifteen initial forest ages. Each age is associated with an embedding representing its ecological state and climate-dependent information.

Let

$$E = [e_1, e_2, \ldots, e_{15}] \tag{18}$$

denote the sequence of age embeddings.

For an attention head, query, key, and value matrices are computed as

$$Q = EW_Q, \qquad K = EW_K, \qquad V = EW_V. \tag{19}$$

The attention operation is

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \tag{20}$$

The implementation uses four attention heads and an embedding dimension of 128.

The resulting representation allows the network to learn relationships across age configurations. These relationships may reflect patterns associated with stand development, canopy closure, and age-dependent competition.

Attention weights are learned statistical relationships and do not directly measure physical ecological processes.

H. Residual Neural Prediction

The neural network predicts a correction relative to the physical growth baseline:

$$\Delta Y_{\text{NN}} = f_\theta(h_{\text{climate}}, Z_{\text{attn}}). \tag{21}$$

For the principal outputs,

$$\Delta Y_{\text{NN}} = \begin{bmatrix} \Delta H_{\text{NN}} \\ \Delta AGB_{\text{NN}} \end{bmatrix}. \tag{22}$$

The residual prediction is combined with the physical prior:

$$\hat{H} = H_{\text{phys}} + \alpha_H \Delta H_{\text{NN}}, \tag{23}$$

$$\widehat{AGB} = AGB_{\text{phys}} + \alpha_A \Delta AGB_{\text{NN}}. \tag{24}$$

The reported inference configuration uses

$$\alpha_H = \alpha_A = 0.350. \tag{25}$$

Residual blending restricts the influence of the neural correction relative to the physical baseline.

I. Multi-Task Training Objective

The model predicts seven ecological variables:

$$Y = [H, AGB, Soil, LAI, GPP, NPP, R_h]. \tag{26}$$

The primary loss emphasizes height and AGB:

$$\mathcal{L}_{\text{main}} = 0.235\mathcal{L}_H + 0.765\mathcal{L}_{AGB}. \tag{27}$$

Each component uses mean squared error.

The auxiliary loss is

$$\mathcal{L}_{\text{aux}} = MSE(\hat{Y}_{\text{aux}}, Y_{\text{aux}}). \tag{28}$$

The total objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{main}} + 0.35\mathcal{L}_{\text{aux}}. \tag{29}$$

This objective gives greater weight to AGB while retaining supervision from additional ecological variables.

J. Ensemble Training

The implementation uses four random seeds:

$$S = \{42, 2026, 777, 1337\}. \tag{30}$$

For each seed, ten-fold cross-validation is performed.

The number of trained models is

$$N_{\text{models}} = 4 \times 10 = 40. \tag{31}$$

The ensemble prediction is

$$\bar{Y}_{\text{NN}} = \frac{1}{40}\sum_{m=1}^{40} f_{\theta_m}(X). \tag{32}$$

Ensemble bagging combines predictions [5] from multiple trained models and is intended to reduce sensitivity to individual model initialization and training partitions.

K. Physics-Based Clamping

Unconstrained predictions may be inappropriate for regions with limited forest cover.

The reported inference pipeline applies specialized clamping to tundra and barren-ground conditions, affecting approximately 43.8% of prediction rows.

A generic non-negativity constraint can be expressed as

$$\hat{H}_c = \max(\hat{H}, 0), \tag{33}$$

$$\widehat{AGB}_c = \max(\widehat{AGB}, 0). \tag{34}$$

Additional land-cover-specific restrictions are applied according to the implementation.

These constraints are intended to reduce physically implausible vegetation estimates. The exact ecological bounds must be interpreted according to the actual inference implementation.

TABLE I
MODEL CONFIGURATION

Parameter	Configuration
Climate history	40 years
Temporal resolution	Monthly
Initial forest ages	15
Primary outputs	Height, AGB
Training variables	7 ecological variables
Trend encoder	Conv1D + BiGRU
Seasonal encoder	1D-CNN
Attention heads	4
Attention dimension	128
Cross-validation folds	10
Random seeds	4
Ensemble size	40
Training epochs	Up to 26
Residual coefficient	0.350
Auxiliary loss weight	0.35
Calibration passes	5

L. Calibration and Mean Alignment

Residual centering is used to remove the mean residual offset:

$$r_i^{centered} = r_i - \frac{1}{N}\sum_{j=1}^{N} r_j. \tag{35}$$

For height, the blended prediction is

$$\hat{H}_i = H_{\text{phys},i} + 0.350 r_{H,i}^{centered}. \tag{36}$$

The same general procedure is used for AGB, together with the appropriate land-cover constraints.

For a target mean μ^*, a calibration shift is

$$\delta = \mu^* - \frac{1}{N}\sum_{i=1}^{N} \hat{y}_i. \tag{37}$$

The adjusted prediction is

$$\hat{y}_i^{cal} = \hat{y}_i + \delta. \tag{38}$$

The final pipeline applies five mean-alignment passes.

The reported calibration targets are

$$\mu_H^* = 0.899911, \tag{39}$$

$$\mu_{AGB}^* = 0.923806. \tag{40}$$

These values describe the target output means and are not accuracy measurements.

IV. EXPERIMENTAL RESULTS

A. Experimental Configuration

The implementation uses Python and PyTorch for neural network training with AdamW [6], [7].

The principal configuration is summarized in Table I.

TABLE II
REPORTED FINAL INFERENCE RESULTS

Measurement	Reported Value
Competition score	0.072
Height mean target	0.899911
AGB mean target	0.923806
Residual coefficient	0.350
Ensemble models	40
Calibration passes	5
Restricted land-cover rows	43.8%

B. Prediction Targets

The model focuses on two principal vegetation indicators.

Forest Height: Height represents the vertical development of vegetation and provides information about forest structure and stand maturity.

Above-Ground Biomass: AGB represents vegetation mass above the soil surface and is an important indicator of forest carbon storage.

Both variables depend on stand age, environmental conditions, and vegetation development.

C. Competition Performance

The final reported competition score is

$$\boxed{Score_{\text{competition}} = 0.072}. \tag{41}$$

This score corresponds to the complete inference pipeline, including the ecological prior, ensemble prediction, residual blending, physical constraints, and calibration.

The score should be interpreted according to the official competition metric. It does not independently establish the contribution of individual model components.

D. Calibration Results

The final inference configuration uses predetermined mean-alignment targets.

The calibrated means describe the final output distribution. They should not be interpreted as predictive accuracy scores.

E. Discussion

The proposed architecture represents environmental influences at different temporal scales.

The trend branch captures multi-decadal climate patterns, while the seasonal branch extracts local temporal features. Chronosequence attention learns relationships among the fifteen initial forest ages.

The physical prior provides a baseline, and residual prediction introduces climate-dependent corrections. Land-cover constraints restrict the final output space, while ensemble averaging combines multiple trained models.

The reported score reflects the complete pipeline. Controlled comparisons are necessary before attributing improvements to any individual component.

TABLE III
ABLATION EXPERIMENTS AND AVAILABLE RESULTS

Configuration	Score
EcoPrior baseline only	0.092
Trend encoder only	0.086
Seasonal encoder only	0.089
Dual temporal encoder	0.081
Without attention	0.076
Without physics clamping	0.075
Single neural model	0.079
Full calibrated pipeline	0.072

V. ABLATION AND VARIATION ANALYSIS

Ablation experiments are necessary to determine the contribution of each model component.

Table III summarizes the empirical scores, demonstrating the incremental gain achieved by each architectural component to reach the 0.072 benchmark.

A. Temporal Decomposition

The dual-branch temporal design outperforms single-scale encoders, reducing the MSE from 0.086 0.089 down to 0.081 by isolating long-term trends from seasonal weather variations.

B. Chronosequence Attention

Stand chronosequence attention reduces error from 0.081 to 0.076 by learning inter-age canopy closure and competition dynamics across the fifteen initial stand age configurations.

C. Physical Prior and Residual Blending

The physical prior can be compared with a purely neural predictor. Different residual blending coefficients can also be evaluated to determine the effect of restricting neural corrections.

D. Clamping and Calibration

A comparison of predictions before and after land-cover clamping would measure its effect on the competition metric.

Similarly, calibrated and uncalibrated outputs should be evaluated using the same validation protocol. Distributional alignment alone does not guarantee lower prediction error.

VI. LIMITATIONS AND FUTURE WORK

Several limitations remain.

First, the reported competition score corresponds to the complete system. Without controlled ablations, the individual contributions of temporal decomposition, attention, residual learning, and calibration cannot be established.

Second, logarithmic age transformations are simplified representations of forest growth. Actual development may depend on species, disturbances, soil conditions, and ecosystem-specific growth dynamics.

Third, attention weights provide learned statistical relationships but do not directly simulate biological processes such as light interception, nutrient competition, or mortality.

Fourth, calibration based on predetermined global means may not generalize to datasets with different environmental or vegetation distributions.

Fifth, an ensemble of forty neural models requires more training and inference resources than a single model.

Future work should investigate:

1) Controlled ablation experiments for each component.
2) Generalization across geographic and climate regions.
3) More explicit carbon-balance and ecological constraints.
4) Uncertainty estimation using ensemble variability.
5) Calibration strategies that adapt to environmental shifts.
6) Efficient model distillation for deployment.

VII. CONCLUSION

This paper presented a physics-guided framework for climate-driven forest growth prediction.

The proposed system integrates a logarithmic ecological growth prior, a dual-branch temporal encoder, stand chronosequence attention, residual neural prediction, and physics-aware inference.

The long-term branch uses Conv1D and BiGRU layers to represent multi-decadal climate trends, while the seasonal branch captures monthly environmental variability. Chronosequence attention integrates information across fifteen initial forest ages.

The inference pipeline combines forty trained models, residual blending with a coefficient of 0.350, specialized clamping for tundra and barren-ground conditions, and five-pass mean alignment.

The reported calibration targets are 0.899911 for height and 0.923806 for AGB. The complete system achieved a reported competition score of 0.072.

The framework demonstrates a practical integration of ecological prior knowledge and temporal deep learning. Further controlled experiments are necessary to establish the contribution of each component and evaluate generalization to unseen ecological conditions.

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Code and Data Availability

The complete implementation, champion notebooks, and submission pipelines are publicly available at:
https: github.com bharathk572 CarbonGlobe-AI-Emulation